Ye Rongyang

+1 (213) 756 1147 · yerongyang@g.ucla.edu . github.com/SnorlaxPapa . LinkedIn

EDUCATION

University of California, Los Angeles (UCLA) - B.S. in Computer Science — Expected Jun 2030

TECHNICAL SKILLS

Languages: Python, JavaScript
Frameworks/Libraries: Triton, PyTorch, Hugging Face, vLLM, Ultralytics
Tools/Technologies: Git, Linux, Excel, Docker

PROJECTS

Mini-vLLM Inference Engine — **vLLM, PyTorch, Triton, Hugging Face**

- Engineered a PyTorch inference runtime for Qwen2.5-3B-Instruct featuring iteration-level scheduling, chunked prefill, hash-based prefix caching, paged KV-cache management, preemption, and CUDA graph capture and replay for decoding.
- Reimplemented the 3B-parameter model architecture in PyTorch and developed a meta-device Safetensors loader with load-time fusion of QKV and SwiGLU gate/up weights, reducing memory overhead and projection kernel launches.
- Built a Triton kernel for slot-mapped KV-cache writes and integrated FlashAttention-2 for variable-length prefill and paged decoding; achieved up to 2.8K output tokens/s on an NVIDIA A100—within 10% of vLLM throughput and 3–7× faster than Hugging Face Transformers.

Custom Triton Kernels — **Triton**

- Built custom Softmax, Layer Normalization, and FlashAttention-2 forward and backward kernels in Triton.
- Implemented causal masking, fused gradient computation, and staged parameter-gradient accumulation.
- Benchmarked on an NVIDIA RTX 4080, matching torch.compile forward throughput and achieving up to 1.5× higher backward throughput.

Object detection model for recyclable materials — **Ultralytics, Roboflow, ONNX**

- Curated a 20,000-image dataset sourced from Roboflow Universe and Kaggle and carried out data cleaning and augmentation via Roboflow.
- Fine-tuned YOLOv8n with Ultralytics, achieving 0.60 mAP@50:95 on the test set; quantized and deployed the model with ONNX Runtime on a Raspberry Pi 4, achieving 2 FPS of on-device inference.

PROFESSIONAL EXPERIENCE

Singapore Armed Forces

Administrative In-Charge — **Jan 2024 - Dec 2025**

- Managed manpower tracking, safety processes, and personnel documentation for a military company of 200+ personnel.
- Built Google Forms and Pandas data-validation workflows and Google Sheets dashboards for administrative data covering 200+ personnel, reducing manual processing by approximately 90%.
- Awarded Commanding Officer’s coin for digitalization efforts.